INTRODUCTION

This procedure has been written to provide guideline information to help the field engineer install the Dry Low NOx system on an MS6001 natural gas fuel machine. This procedure was written for a unit that shipped without the combustion system installed and for a unit that is going to be retrofitted with the old combustion system removed.

DESIGN

The Dry Low NOx system consists of the combustion system, fuel gas system, inlet bleed heat system, cooling and sealing air system and air extraction system, cooling water system, piping interconnection, and additional transducers installed in the accessory compartment.

REFERENCE DOCUMENTS

No.	Description
MLI 0719	Case Arrangement - Combustion - Outer
MLI 0991	Piping Arrangement - Gas Fuel Module
MLI 0701	Chamber Arrangement - Combustion
MLI 0703	Cap and Liner Arrangement - Combustion
MLI 1127	Conduit Mod Arrangement - Flame Detectors
MLI 0915	Piping Arrangement - Cooling Water
MLI 0963	Piping Arrangement - Cooling Water - Turbine
MLI 1104	Conduit Arrangement - Access Compartment Module
MLI 0557	Transducer Arrangement
MLI 0969	Piping Arrangement - Base Interconnect
MLI 0962	Piping Arrangement - Primary Gas
MLI 0962	Piping Arrangement - Secondary Gas
MLI 0962	Piping Arrangement - Tertiary Gas
MLI 0924	Piping Arrangement - Air Extraction
MLI 0909	Piping Arrangement - Clg and Slg Air
104E8461	Piping Arrangement - Inlet Heating
104E7403	Piping Arrangement - Fuel Nozzle Purge

RECEIVE DRY LOW NOX MODIFICATION

The dry low NOx modification package may ship in one box or may be separated by model list. The complete package should be checked for inventory and damage. The parts list and the packing list should be compared to the modification drawings to ensure that all the parts identified for the modification were shipped.

INSTALLATION OF THE DRY LOW NOX SYSTEM:

For a retrofit unit, the combustion system, cooling and sealing air system, extraction air system, and gas manifolds will have to be removed. In the turbine compartment you will start the Dry Low NOx installation with the compressor casings, compressor discharge casing, and turbine casing intact.

If a new unit arrives without the Dry Low NOx System installed, the unit will have no manifolds, cooling and sealing air piping, combustion system, or extraction piping installed. On a new unit without the combustion system or on a retrofit, the turbine compartment roof will need to be removed.

(MLI 0701) Use part 4, screw, machine pan head to attach flow sleeves to the aft outer combustion case for all of the combustion cases, Items 1 and 5. (See Figure 1).

(MLI 0719) Install assembly of outer combustion case and flow sleeve to the compressor discharge case (MLI 0805) using pt. 5 gasket. See Figures 2 and 3 for bolts, and the bolting and doweling arrangement (MLI 1604).

Outer surface of outer crossfire tube (pt. 15) should be smooth and free of nicks and burrs before insertion through packing rings. Install outer crossfire tube (pt. 15) to outer combustion case (MLI 0719) using pts. 16 through 19. (See Figure 4.)

Do not re-use packing rings (pt. 16). Compress packing rings until gap between seal ring (pt. 17) and combustion case boss is a constant 0.080 + 0.015 all around, see section A-6 in Figure 4. This will result in a total packing ring compression of approximately 0.150.

Prior to installing each combustion liner, install crossfire tube halves (pt. 13,14) into outer combustion case crossfire holes as far as they will go. Orient crossfire tube halves so that pt. 13 is always C.W. and pt. 14 C.C.W. of chamber centerline as shown in section A-6 of Figure 4. The tube halves can be held in the above position with pt. 12 retainers partially installed over the small diameter end of tubes. When crossfire tube halves are in place, insert combustion liner until combustion case and liner stops are fully engaged.

Disengage each pt. 12 retainer from crossfire collar until slot on tube lines up with retainer. Fully engage retainer pt. 12 with tube slots until retainer is in contact with stop pin (see sec. A-1, Figure 4). With retainer fully engaged, retainer end to be 0.434" + 0.100" from face of outer combustion case. See Section A-6, Figure 4.

To install spark plug assembly **(MLI 1214)** (Figure 5), remove igniter assembly from base assembly and install base assembly using pt. 8 gasket and pt. 9 bolts. Insert igniter assembly through base assembly and flow sleeve and penetrate into liner 0.380 from liner O.D. Torque swagelok nut of base assembly 1.25 turns to fix igniter position. To retighten, hand tighten, then torque 0.25 turn. All bolts to be torqued per Item 2, Sh. 1 unless otherwise specified.

To install fuel nozzle/cover assembly, remove secondary nozzle from cover. Installing the cover assembly with secondary nozzle attached will risk damage to cap centerbody assembly. Check the five (5) floating collars on liner cap assembly for free movement. These collars must move to prevent damage during assembly of cover to wrapper.

Note: For exact configuration of cover, see Figures 6, 7 and 8.

Note: Cover orientation in Figure 7, primary gas connections are 72 degrees CCW of chamber centerlines and/or liquid fuel connections are at 12 o'clock.

Engage the three (3) guide pins with holes in combustion case forward flange and move cover towards combustion case. Engage rabbet fit between cover and combustion case. Nozzle tip engagement with floating collars is 0.375 before cover face meets combustion case face. Install and torque cover mounting bolts.

To install secondary nozzle assemblies, note orientation of secondary nozzle gas flanges in Figure 7. Locate gasket pt. 21 in grove of secondary nozzle body flange. Carefully insert nozzle assembly through cover center hole. Tip of nozzle engages feature inside cap centerbody when nozzle flange is approximately 0.500 from cover face. Do not force the secondary nozzle into the centerbody. Hold nozzle body flush to cover face and install mounting bolts pt. 22 to a uniform torque all around progressively torque bolts moving to opposite side of flange for next bolt.

(MLI 0924) The air extraction piping parts and location can be found on drawing 104E8187 (Figures 9 and 10). Install the manifold sections around the turbine section first. Then install the lower section (i.e., item 5, 6, 7) next, working your way out of the turbine section. Referring to piping arrangement inlet heating (drawing 104E8461, Figures 11, 12 and 13), install the support structure for the inlet heating piping from the turbine section to the inlet filter house. When installing the VA-20-1 valve, install so it can be read from the walkway below. This may require rotating the valve. Install the inlet heating piping referring to the drawing.

(MLI 0962) Once the combustion cans, combustion liners, combustion covers, fuel nozzles and air extraction piping have been installed, start setting the tertiary, secondary, and primary gas manifolds into position. It is necessary to have the combustion system installed in order to take measurements for the manifolds.

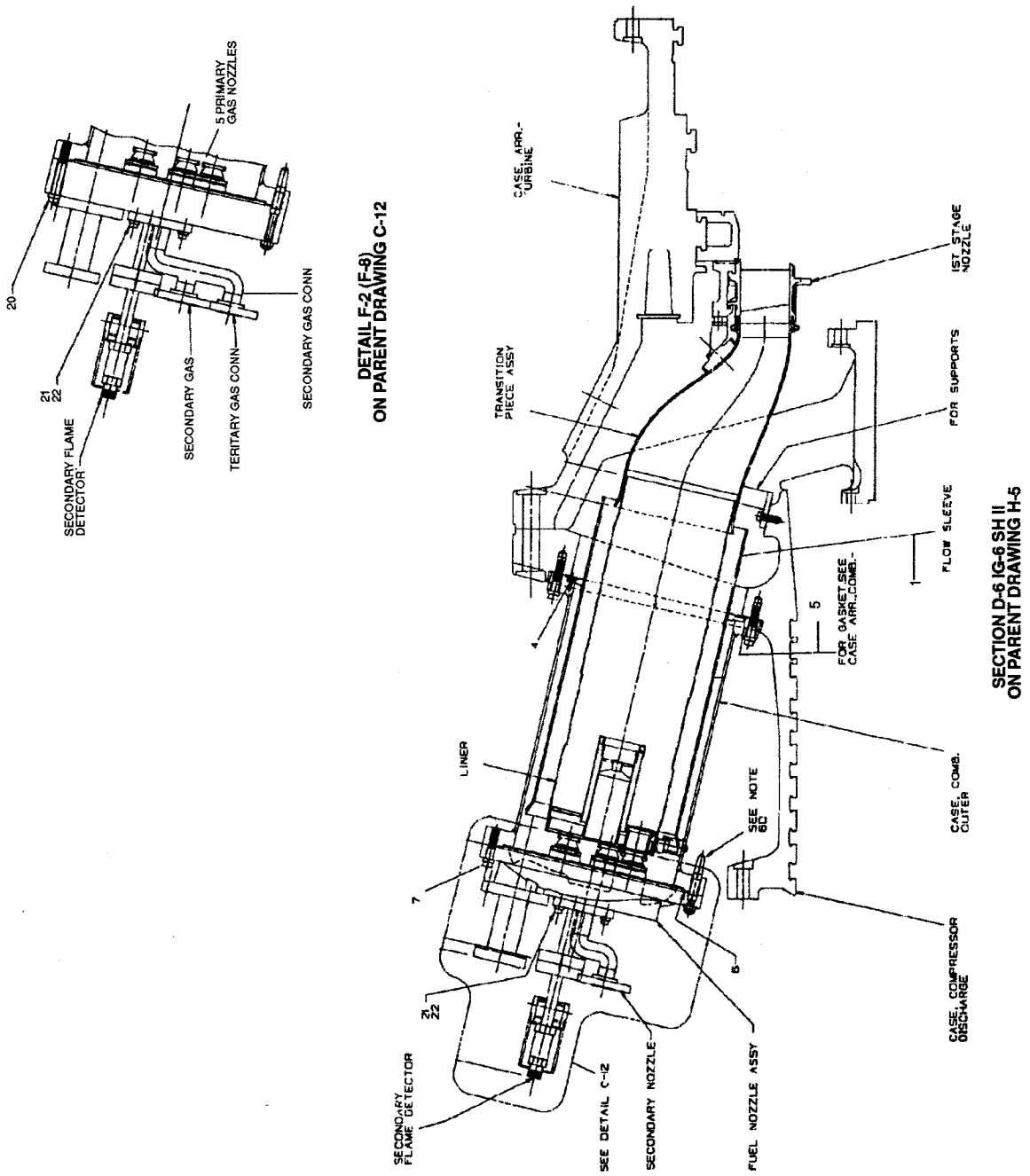


Figure 1. Combustion System Components (MLI 0701) - Typical Layout

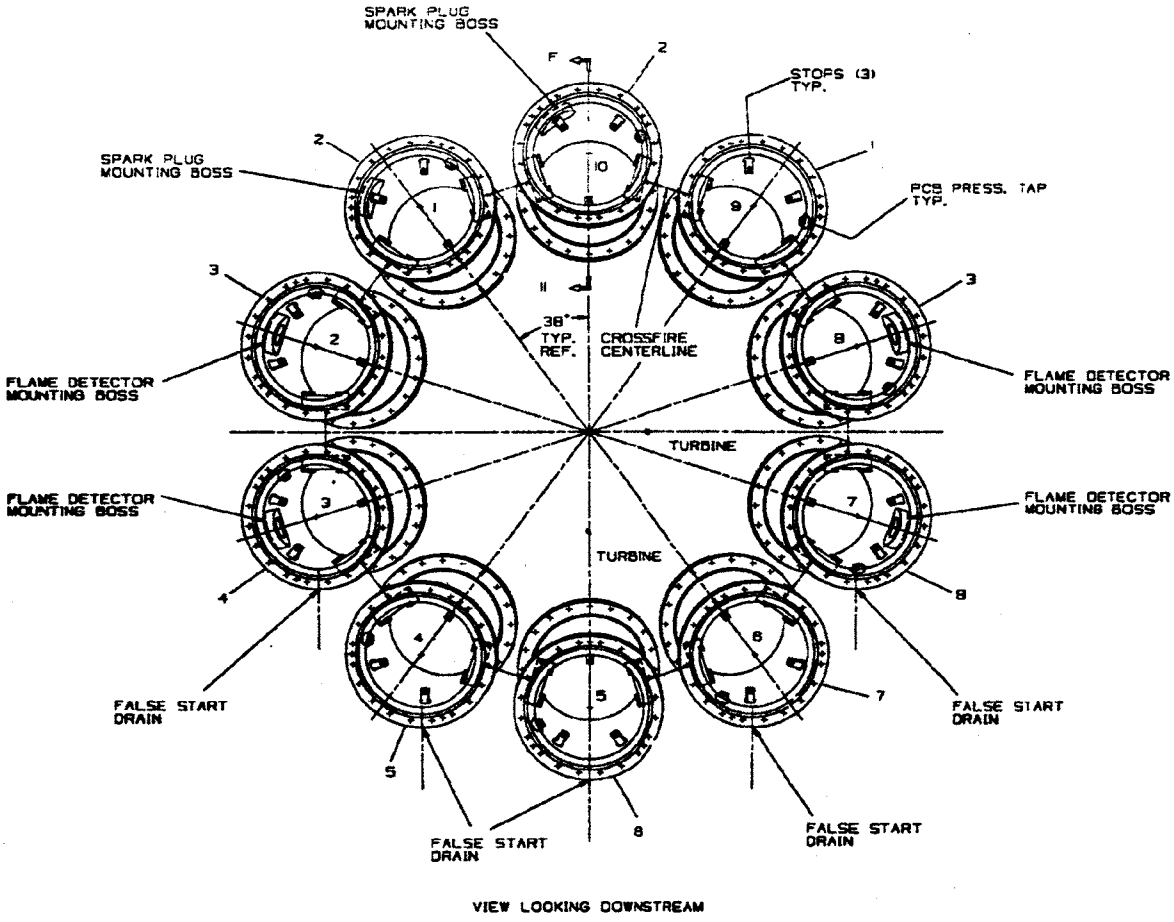


Figure 2. Outer Combustion Casing (MLI 0710) - Typical Layout

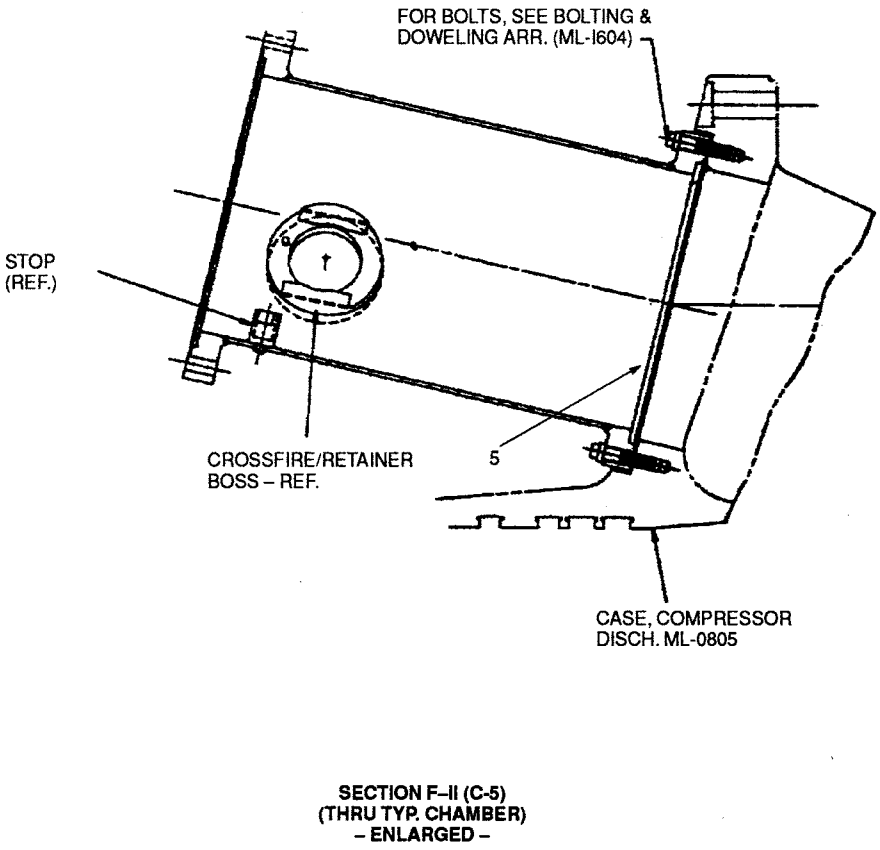


Figure 3. Combustion Outer Casing (MLI 0719) - Typical Layout

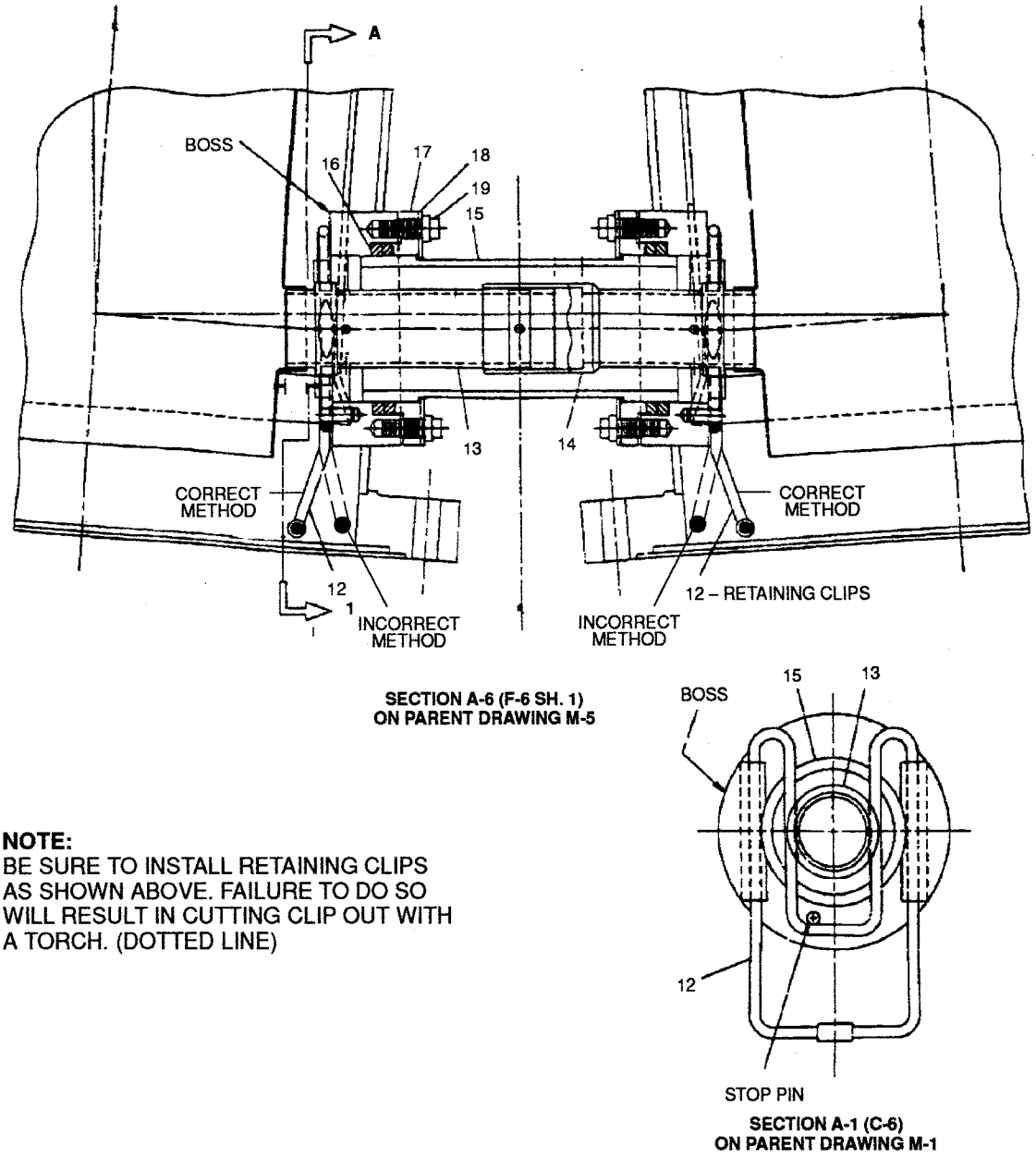


Figure 4. Crossfire Tube Assembly MLI 0701) - Typical Layout

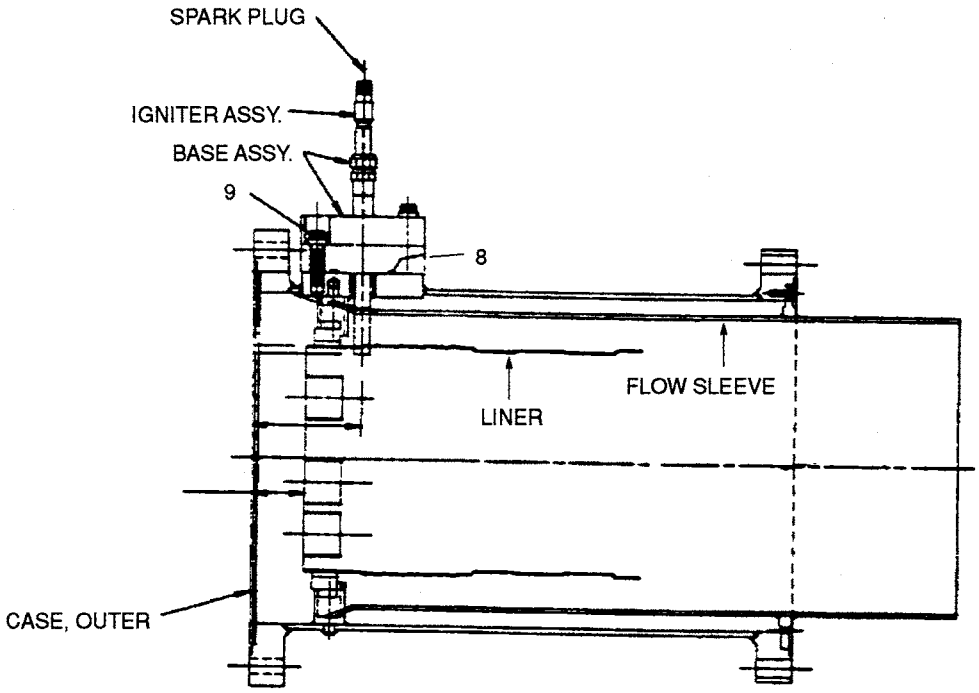


Figure 5. Spark Plug Assembly (MLI 0701) - Typical Layout

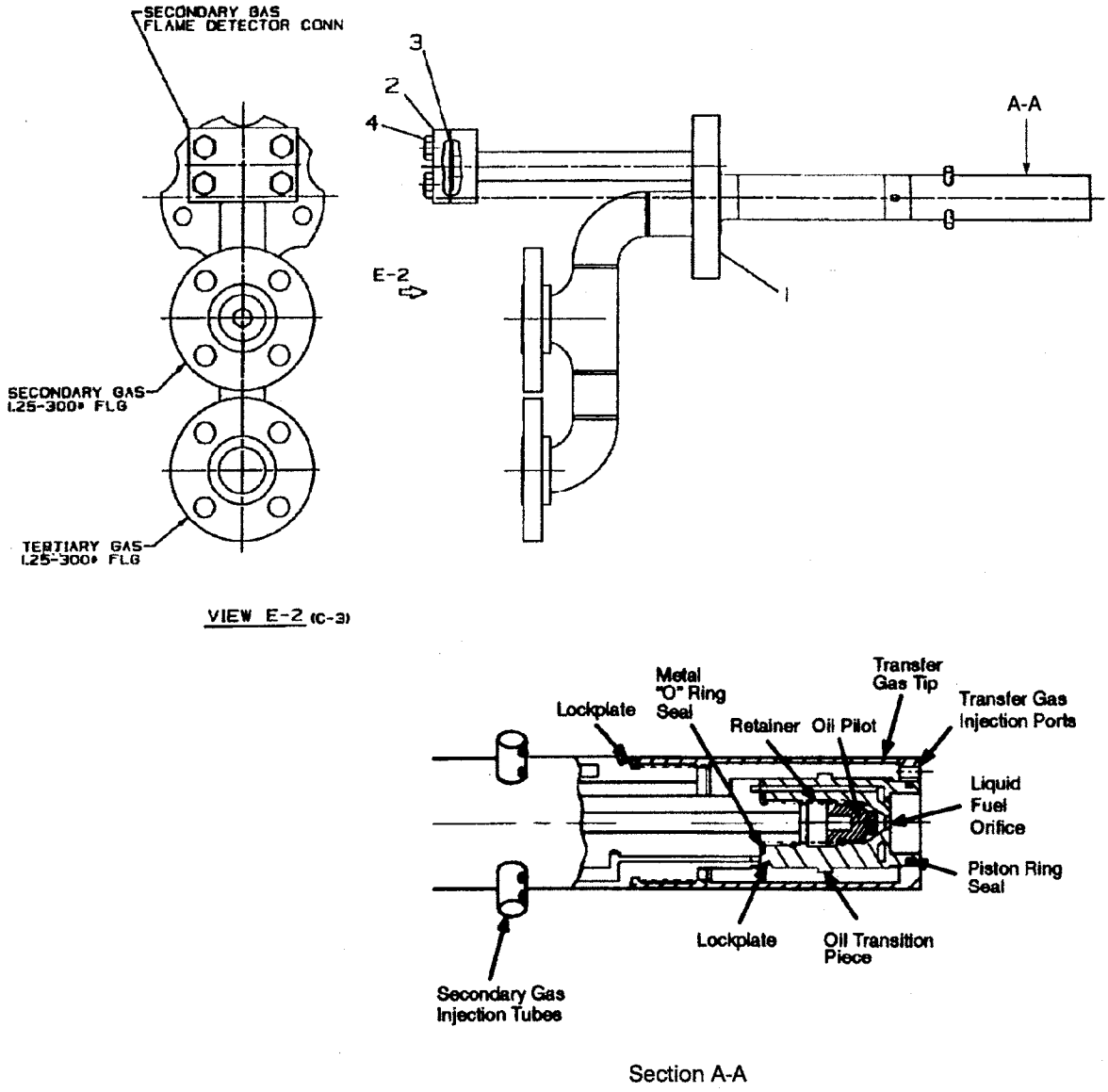


Figure 6. Secondary Fuel Gas Nozzle Assembly (MLI 0566) - Typical Layout

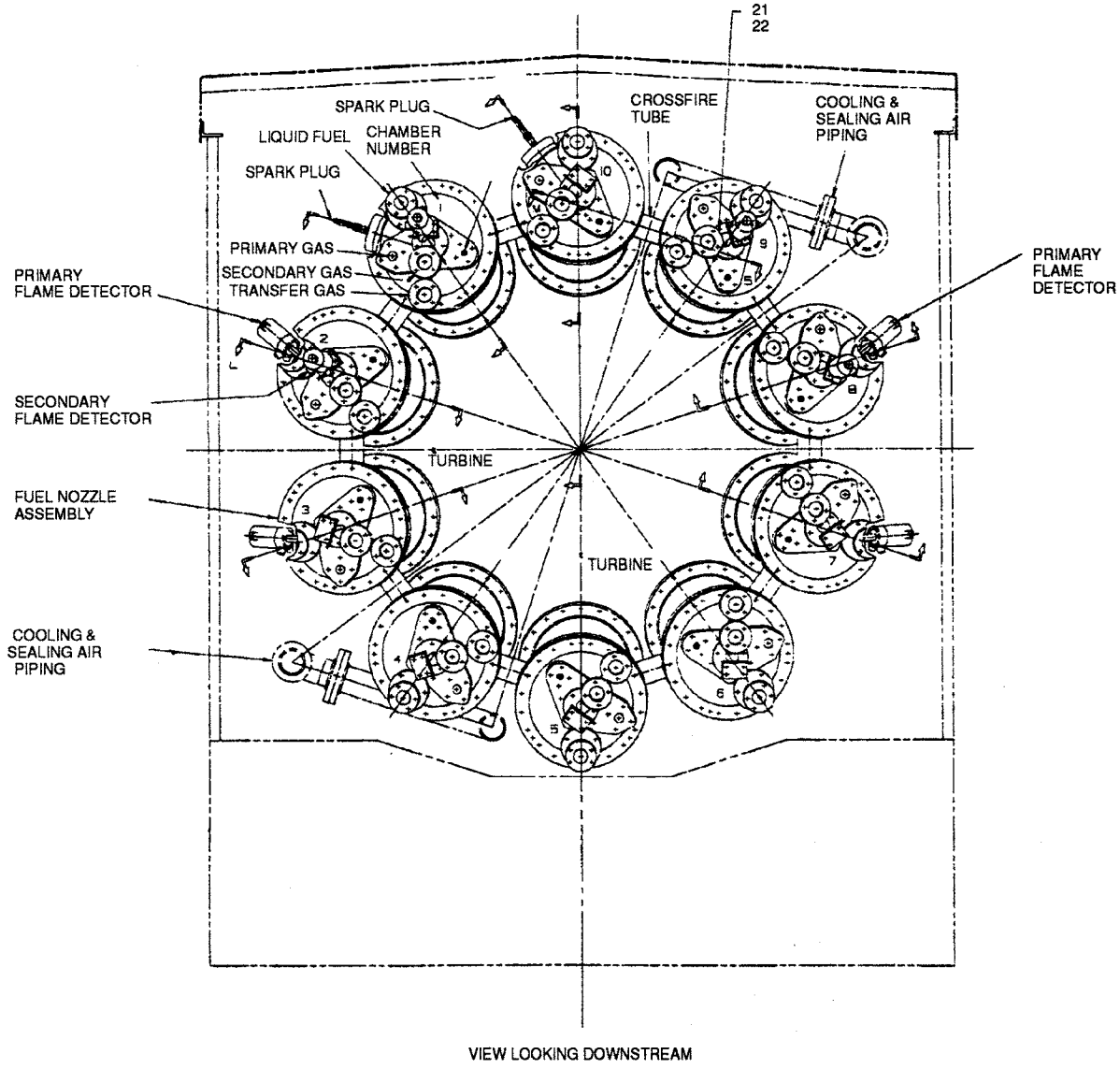


Figure 7. Combustion Chamber Arrangement (MLI 0701) - Typical Layout

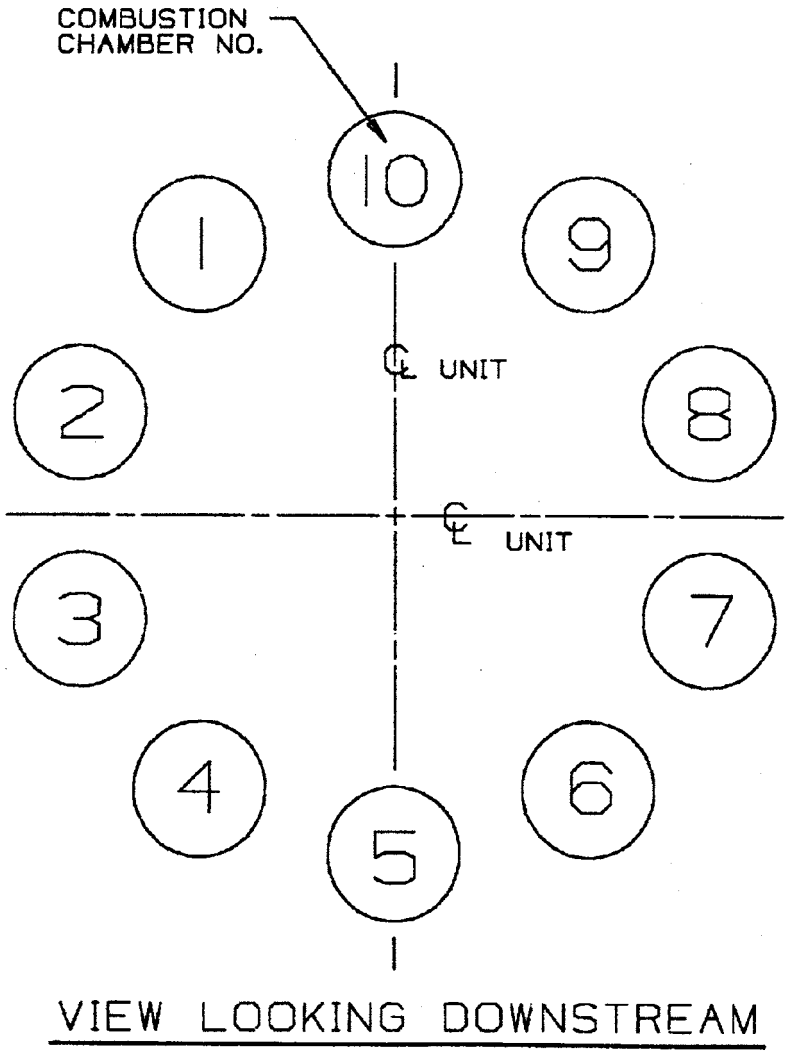


Figure 8. Combustion Cap and Liner Arrangement (MLI 0703) - Typical Layout

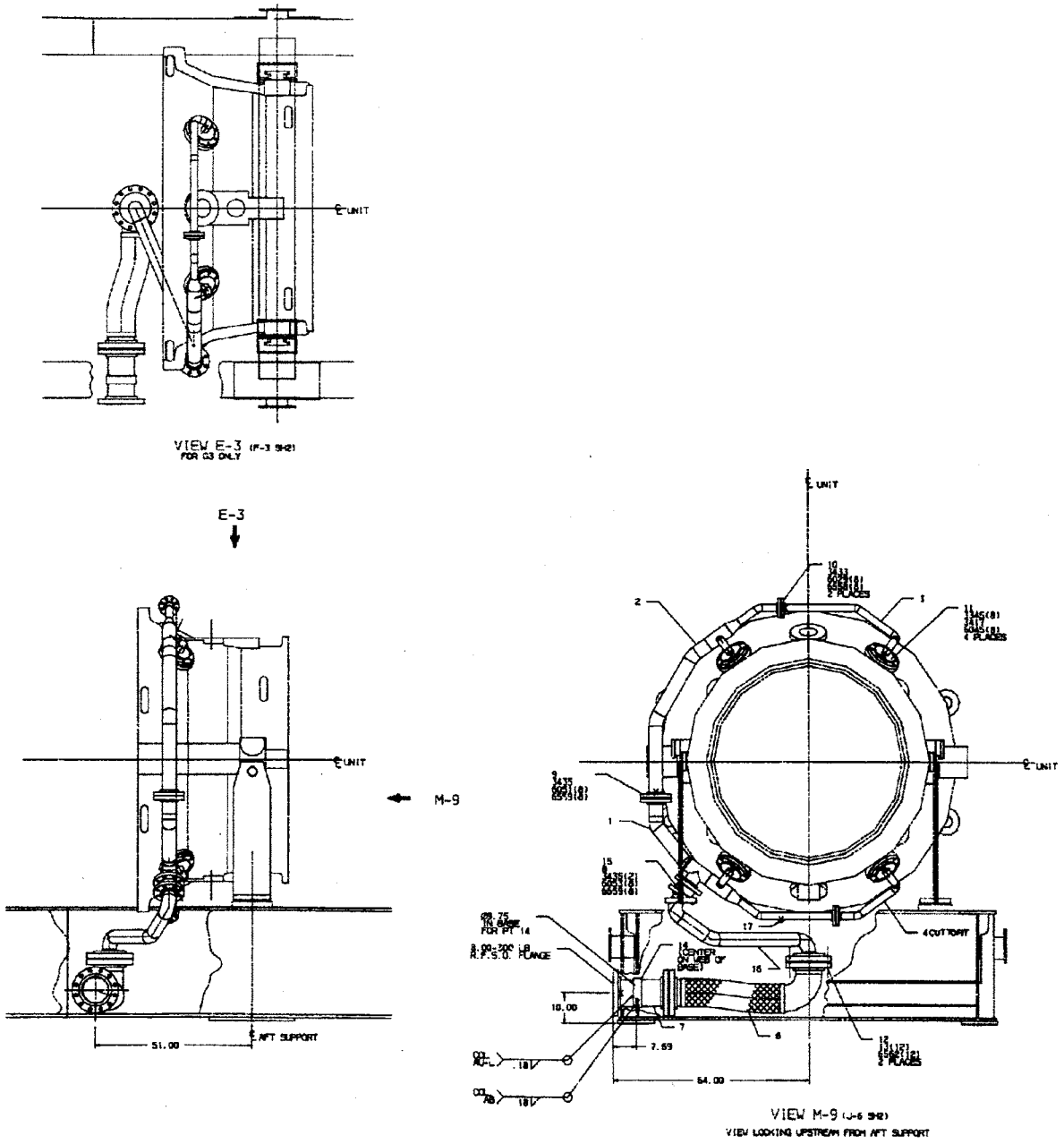


Figure 9. Air Extraction Piping Arrangement (MLI 0924) - Typical Layout

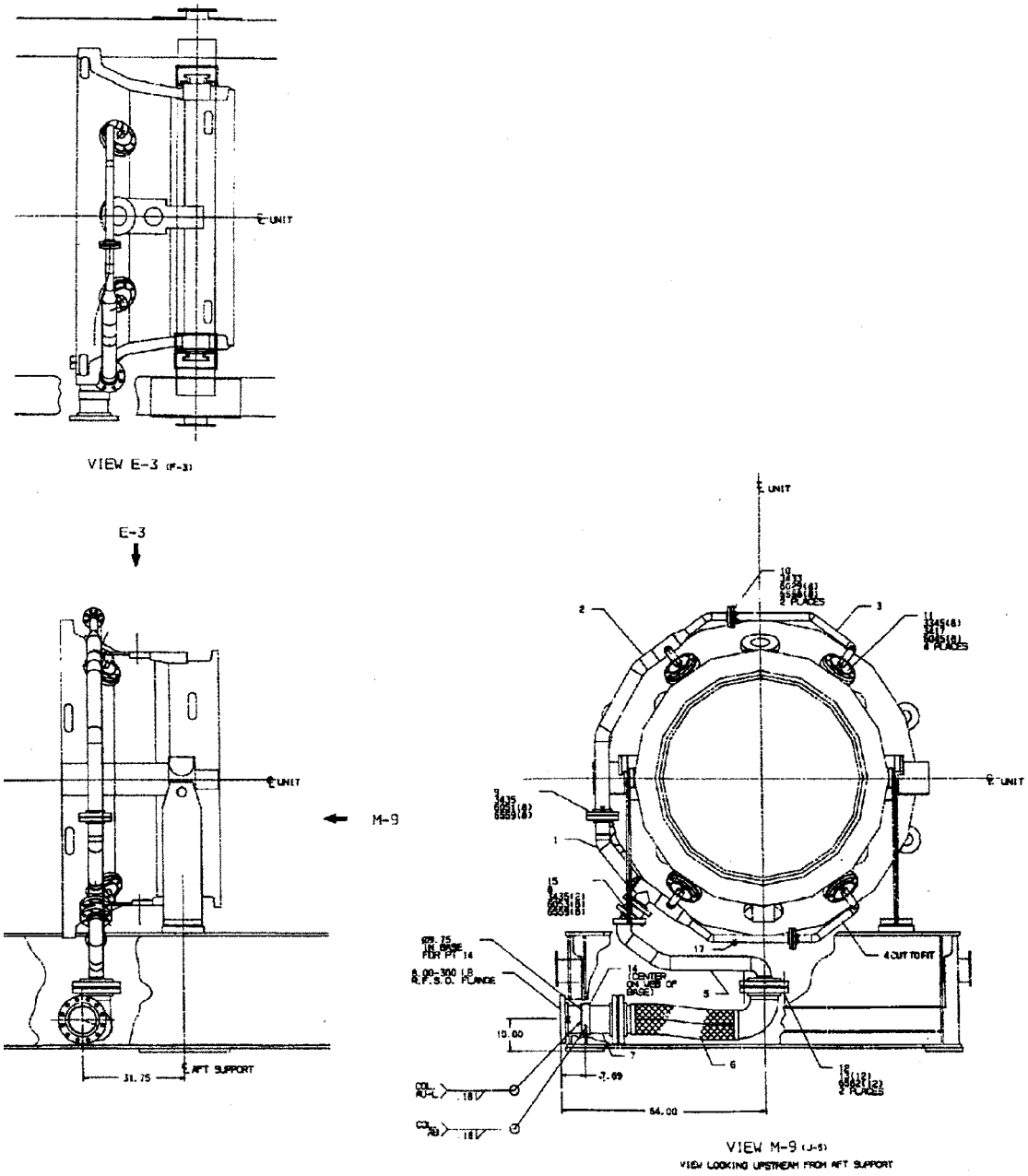


Figure 10. Air Extraction Piping Arrangement (MLI 0924) - Typical Layout

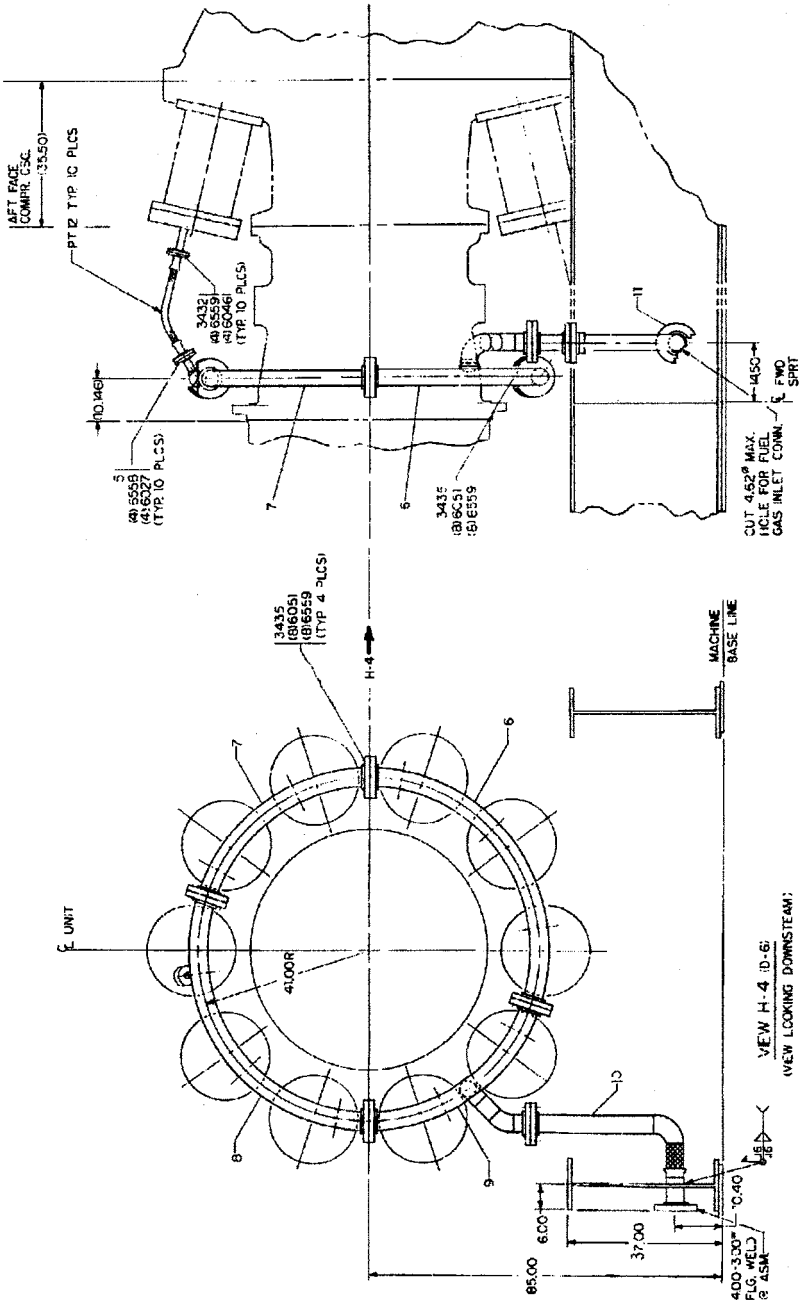


Figure 11. Primary Gas Piping Arrangement (MLI 0962) - Typical Layout

Starting with the tertiary gas manifold, assemble the bottom half and lower into the turbine compartment. Block the lower half into position while assembling the upper half. When the upper half is assembled, assemble the two halves together.

When the tertiary gas manifold is completely assembled support the manifold by blocking and hanging with come-a-longs temporarily if the roof is on. If the roof is not on, block between the upper manifold and the compressor casing for proper positioning of the manifold.

Follow the same procedure for the secondary and primary gas manifolds as for the tertiary gas manifold. With the three manifolds in position supported temporarily, start installing the flex hoses from the manifolds to the fuel nozzles. After all the hoses have been installed, install the main support brackets. These are the brackets mounted on the turbine compartment I-beam. The brackets are a saddle arrangement that are the main support for the manifolds. See Figure 11, and drawings 104D3543, 104D3541 and 104D3542 for dimensions. When the manifolds are secured in position, locate the holes that must be cut in the turbine base I-beam for connections to the gas fuel module piping. See 104D3541, 104D3542 and 104D3543.

(MLI 0909) The cooling and sealing air system can now be installed. This will complete the heavy piping inside the turbine compartment. Install the lower piping that connects to the compressor casing and install the saddle support bracket. Connect the upper piping that connects to the compressor casing (pts. 19, 18 and 32). See Figure 12. Continue connecting the rest of the piping. Do not tighten flanges tight until all of the piping is connected.

When all piping is connected and in position, tighten all flanges. This piping goes through the turbine compartment roof, and therefore it is necessary to cut new holes in the roof. Some of the piping can be installed before the roof is in position to ease in the installation of the piping. Once all piping is installed, install the cowling covers to go around the new piping and to cover the new holes in the roof.

(MLI 0963) When all of the heavy piping inside the turbine compartment has been installed, the cooling water piping and the electrical connections for the flame detectors can be installed.

Note: When installing flame detectors, a "line of sight" check should be done. Look into the primary flame detector hole and see if you can see the primary fuel nozzle.

The cooling water supply for the flame detectors comes from the cooling water header for the turbine support legs (see Figure 13). The flame detectors will have a supply and return to the cooling water header for the support legs. **MLI 0915** (drawing 176D3555) shows detailed connections of the cooling water piping on the flame detectors. **MLI 1127** (drawing 103E5839) shows the conduit arrangement for the flame detector connections. See Figures 14 and 14A.

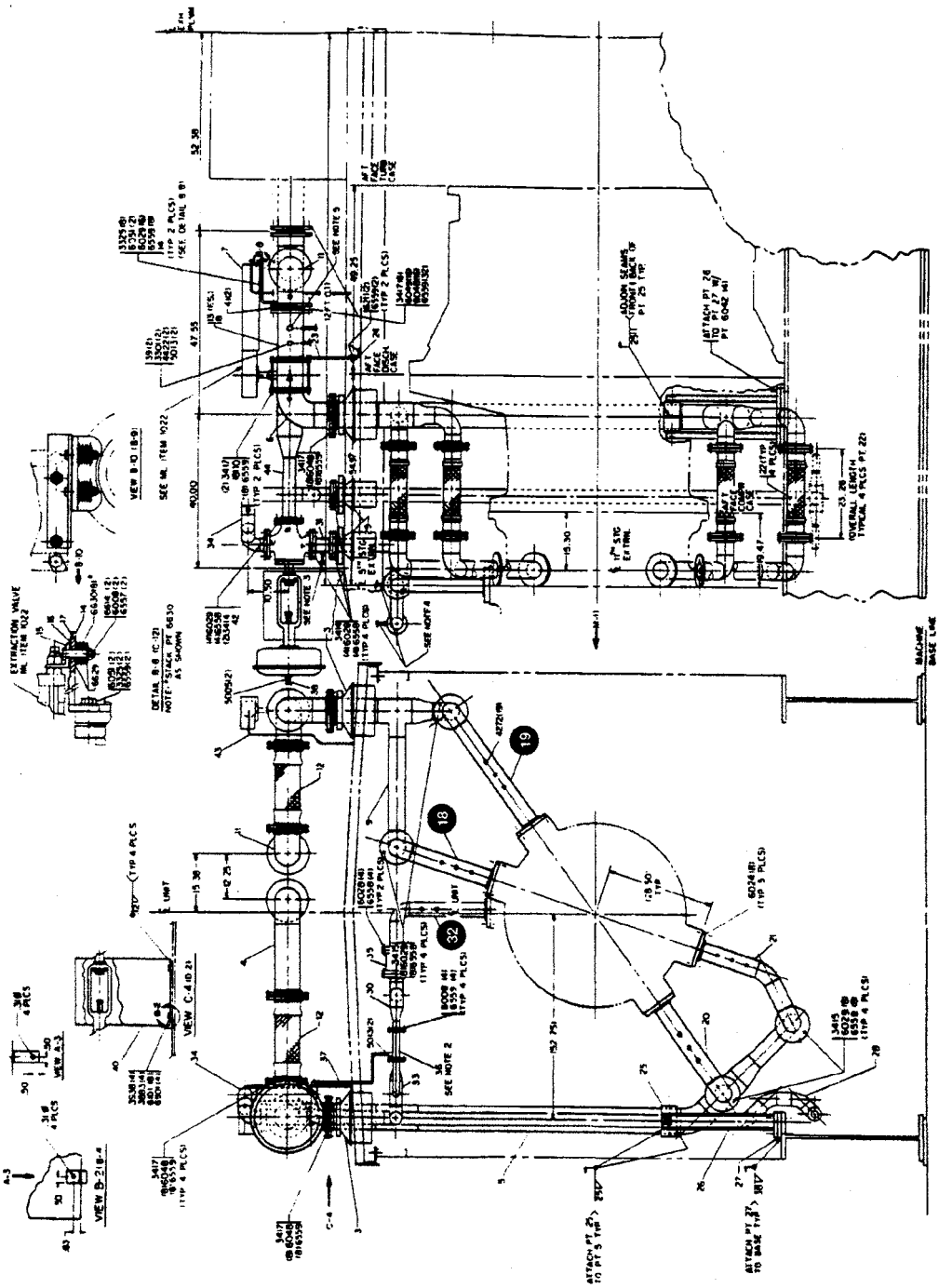


Figure 12. Cooling and Sealing Air Piping Arrangement (MLI 0909) - Typical Layout

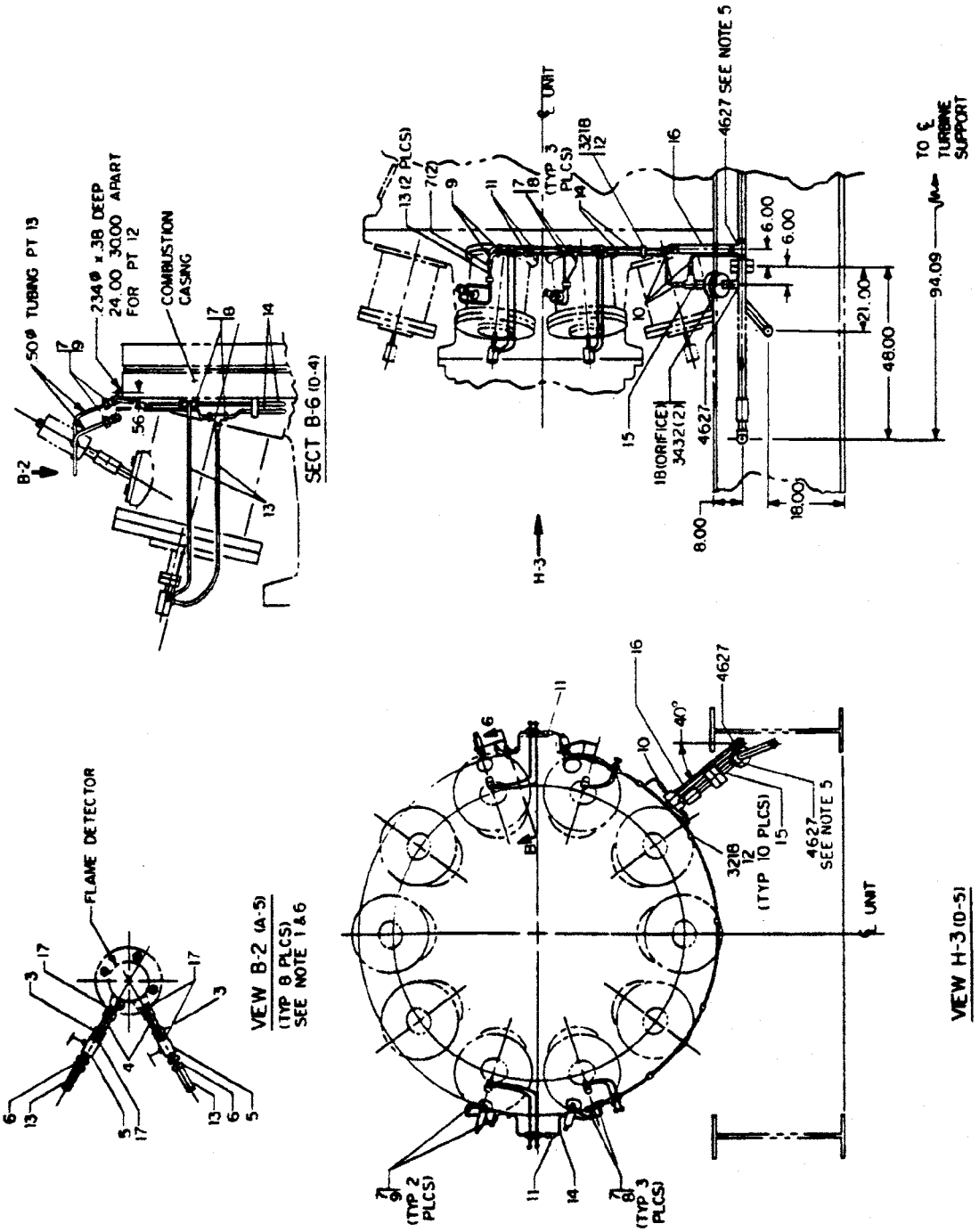


Figure 13. Cooling Water Piping Arrangement (MLI 0915) - Typical Layout

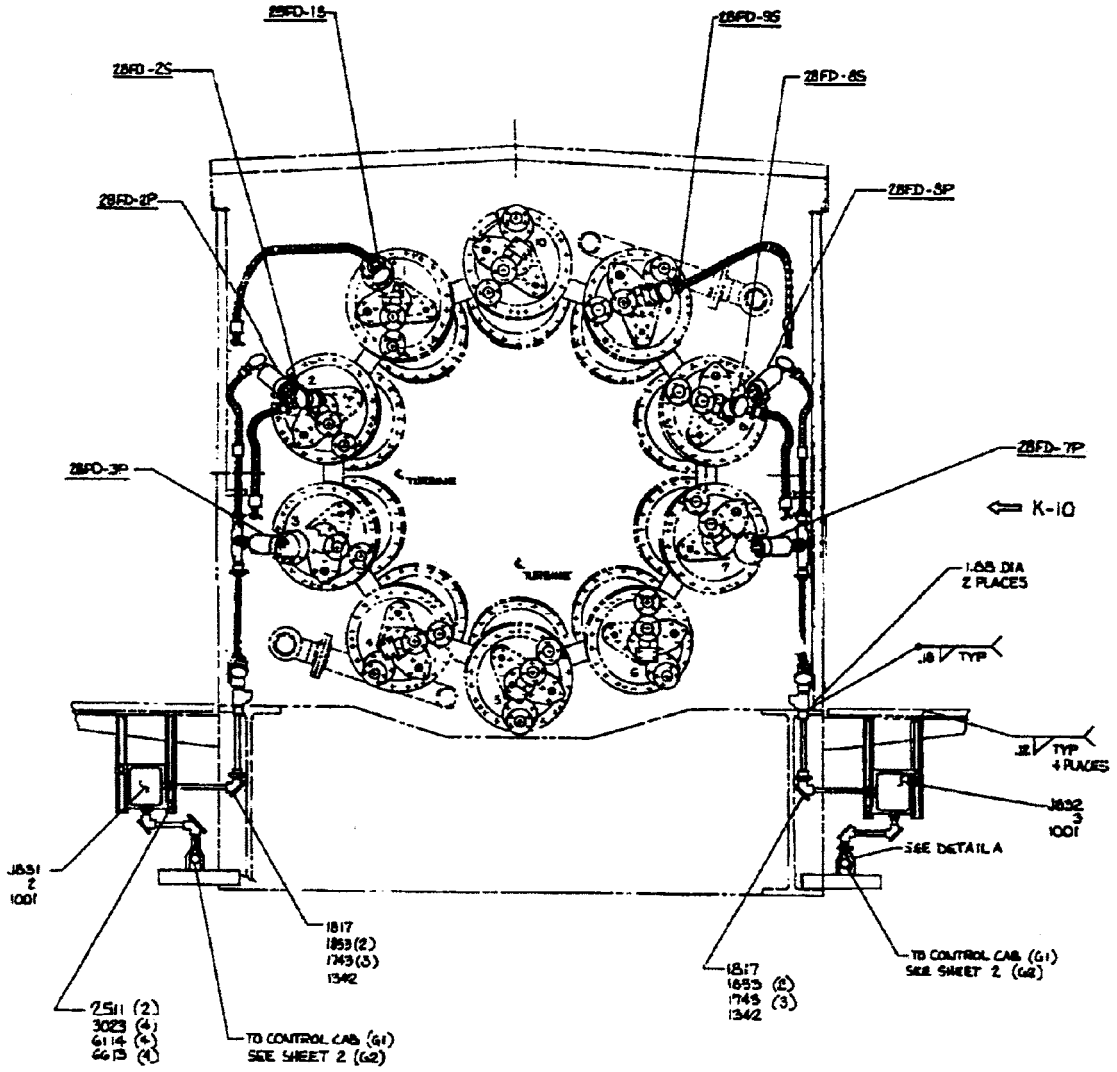


Figure 14. Flame Detectors Conduit Med Arrangement (MLI 1127) - Typical Layout

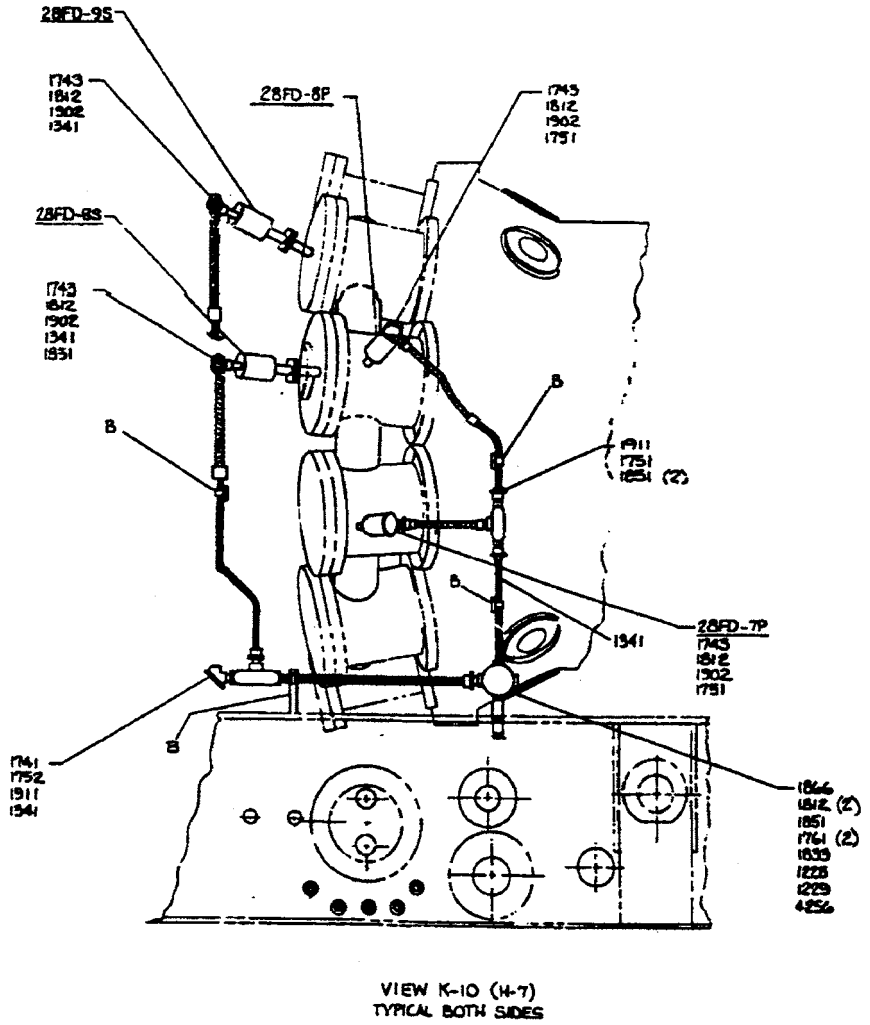


Figure 14A. Flame Detectors Conduit Med Arrangement (MLI 1127) - Typical Layout

MLI 0991 (drawing 104E7403) shows the base interconnection from the fuel gas module to the gas manifolds. The fuel nozzle purge piping can be installed after the gas module is in place and the manifolds have been installed with the piping through the I-beam of the turbine base. Installation of the interconnection piping will be different from site to site because of the different configuration of external piping and conduit runs outside the unit. Therefore, the piping may not be installed exactly to drawing. If you must deviate from the piping drawing, be careful with the pipe layout. Elevation and the number of 45 and 90 degree angles must be considered so as not to create pockets for condensate or liquids to accumulate. See Figures 15, 15A and 16.

Ref. 104E8461. With piping for the air extraction completed and through the I-beam of the turbine base, the inlet heating system can be installed. Erect the support steel and then erect the piping. The VA-20 valve may have to be oriented so the customer operators can read the valve setting when on the walkway. The drawing shows the VA-20 valve under the inlet plenum. This is not feasible because there is not enough vertical clearance from the pipe run and the bottom of the inlet ducting, therefore the valve must be positioned to the right of the inlet plenum. See Figures 17, 17A and 18 for details of the support steel and the piping run.

Ref. MLI 0557 and MLI 1104. The conduit arrangement for the transducers and the flame detectors is straight forward and only requires drawing interpretation.

The turbine compartment space is limited because of all the piping. Therefore, installation of the conduits will be performed last to avoid damage to the conduits while installing the piping.

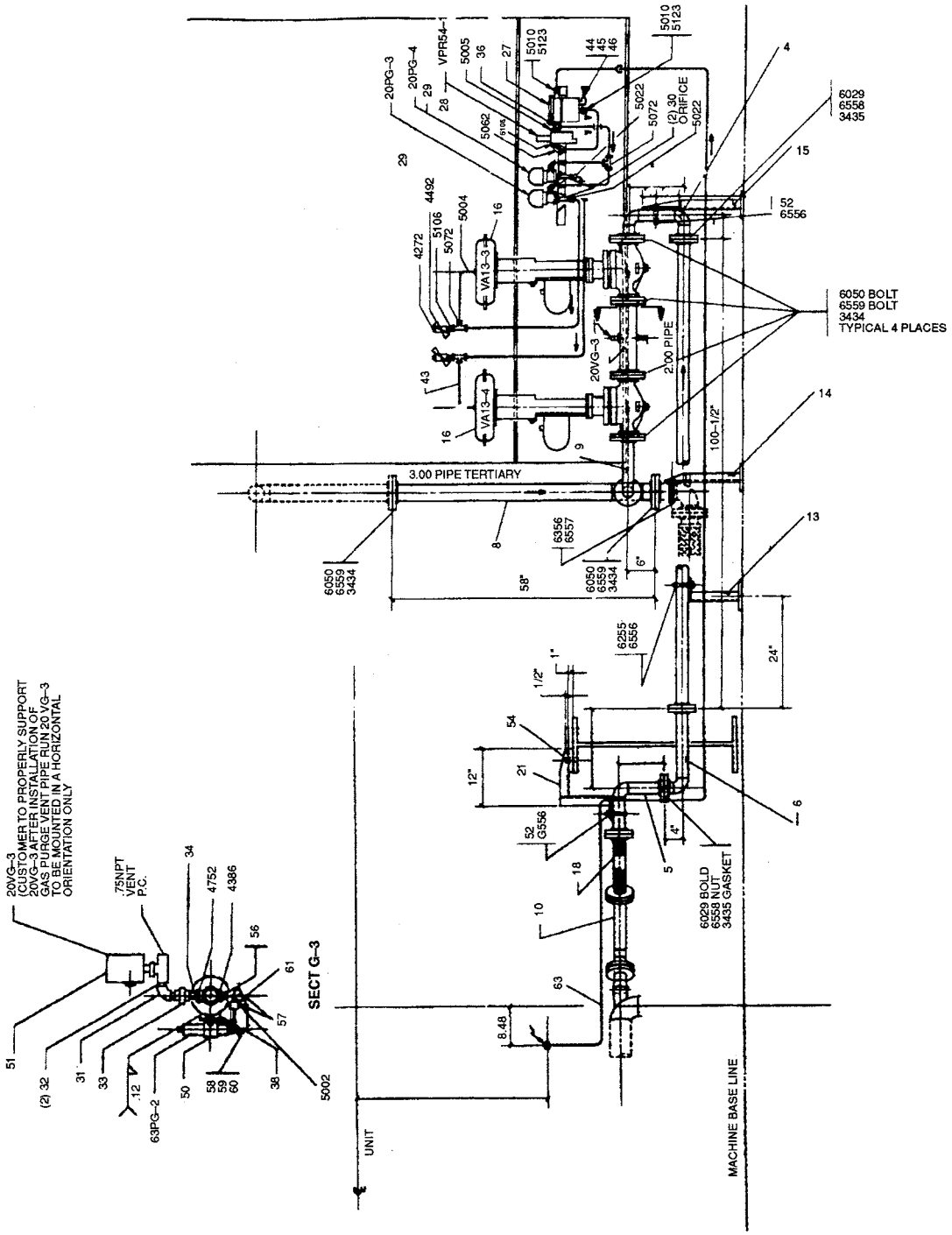


Figure 15. Fuel Nozzle Purge Piping Arrangement (104E7403) - Typical Layout

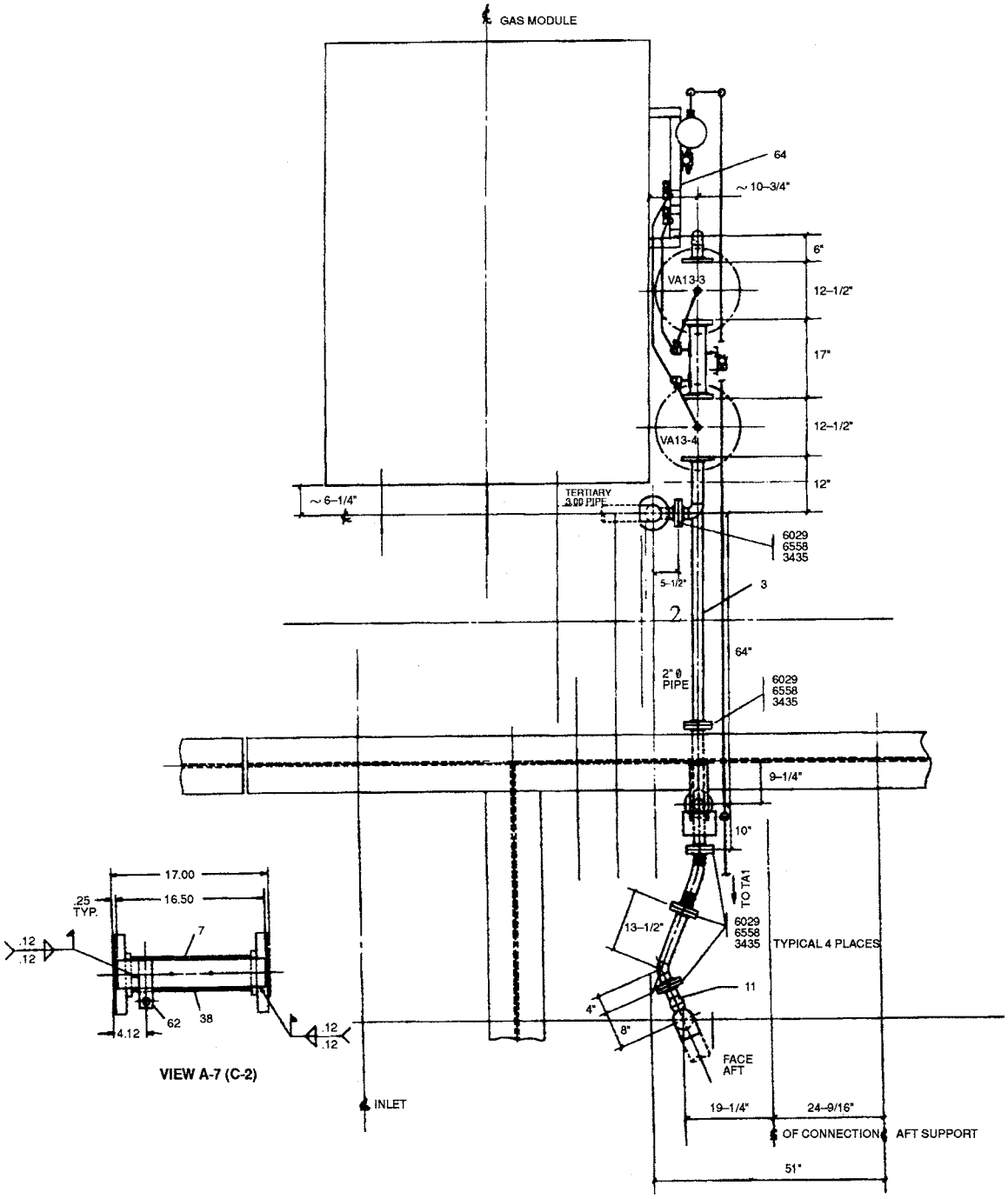


Figure 15A. Fuel Nozzle Purge Piping Arrangement (104E7403) - Typical Layout

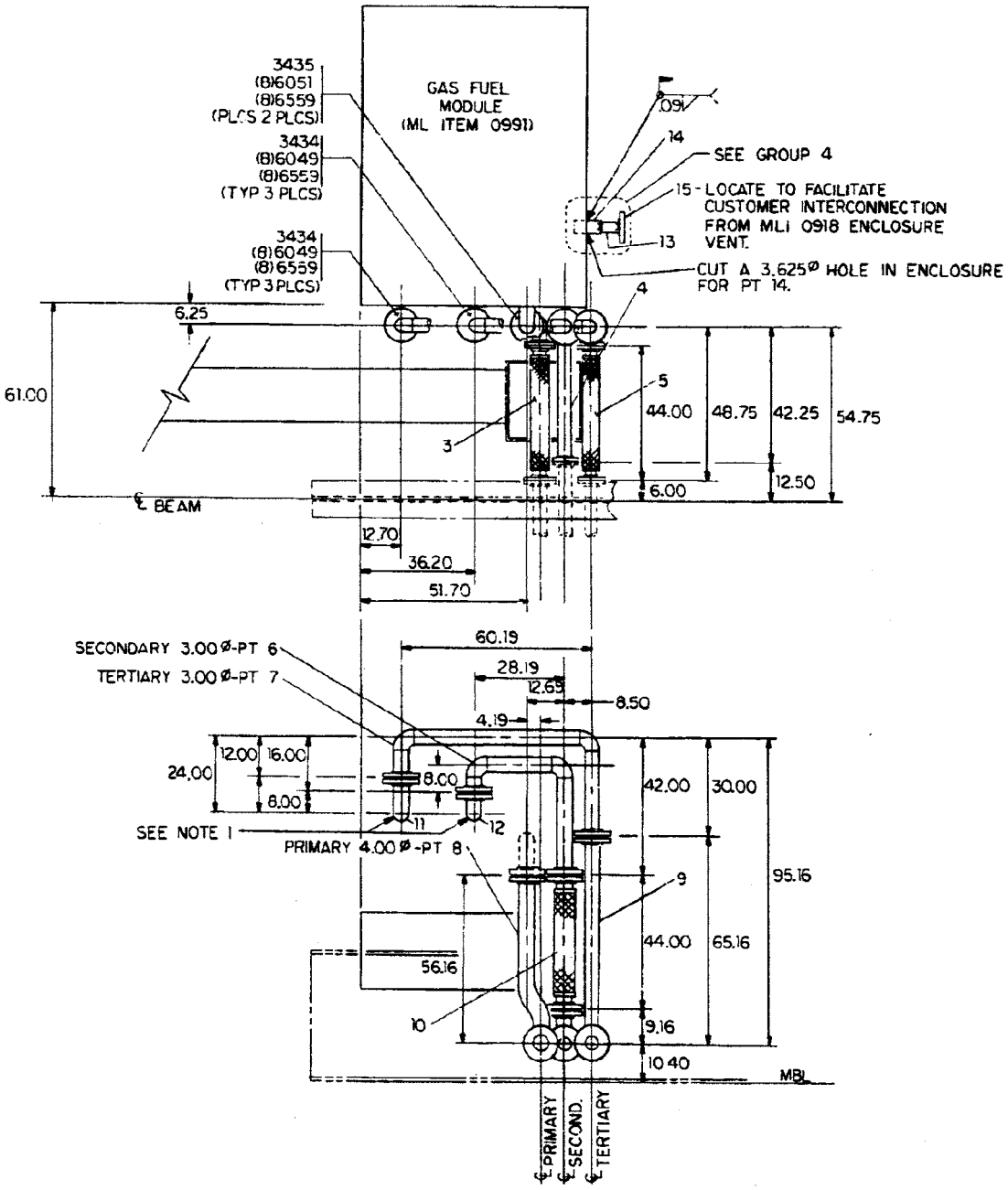


Figure 16. Base Interconnect Piping Arrangement (MLI 0969) - Typical Layout

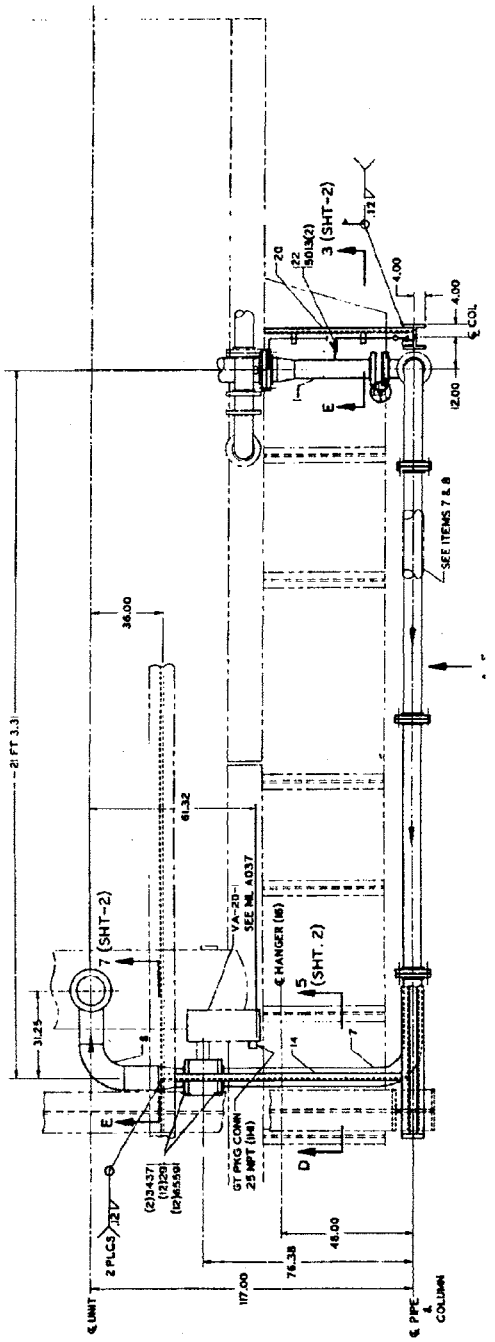


Figure 17. Inlet Heating Piping Arrangement (MLI A053) - Typical Layout

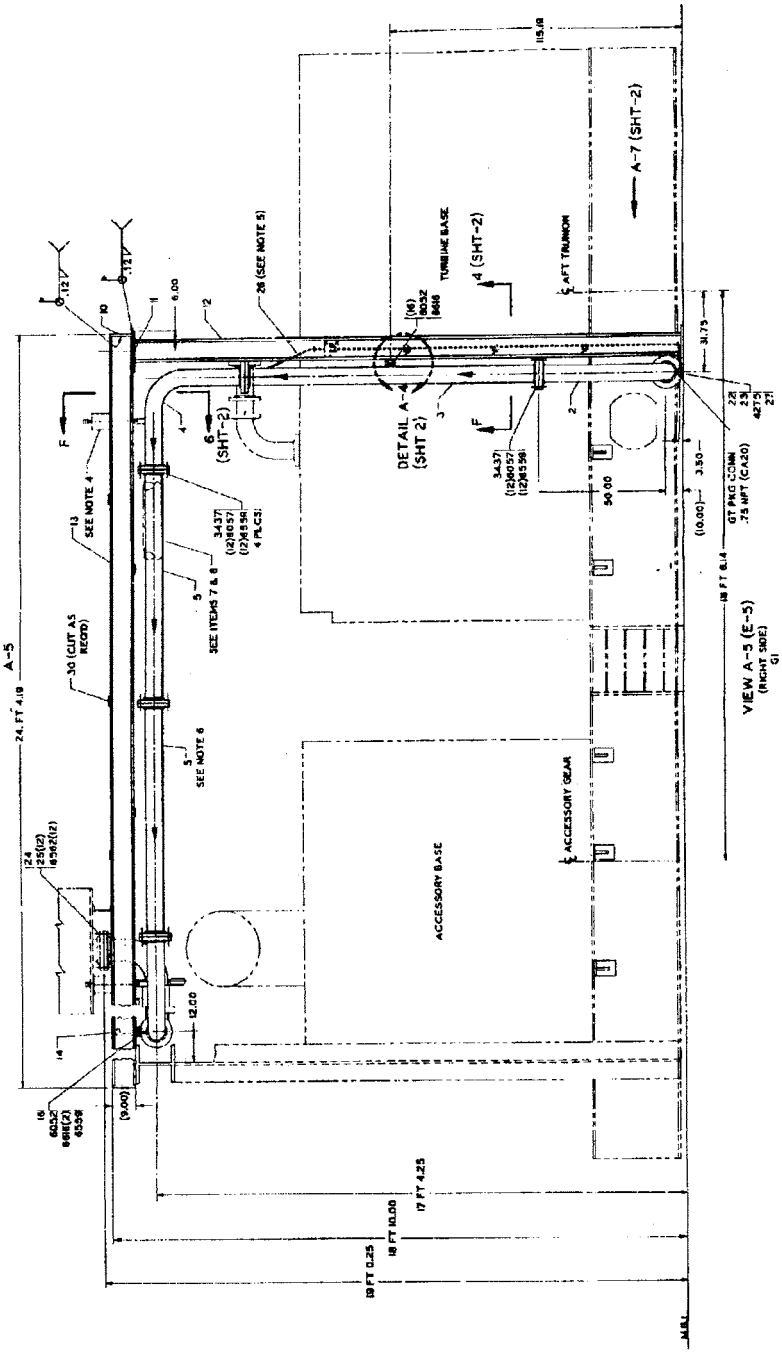


Figure 17A. Inlet Heating Piping Arrangement (MLI A053) - Typical Layout

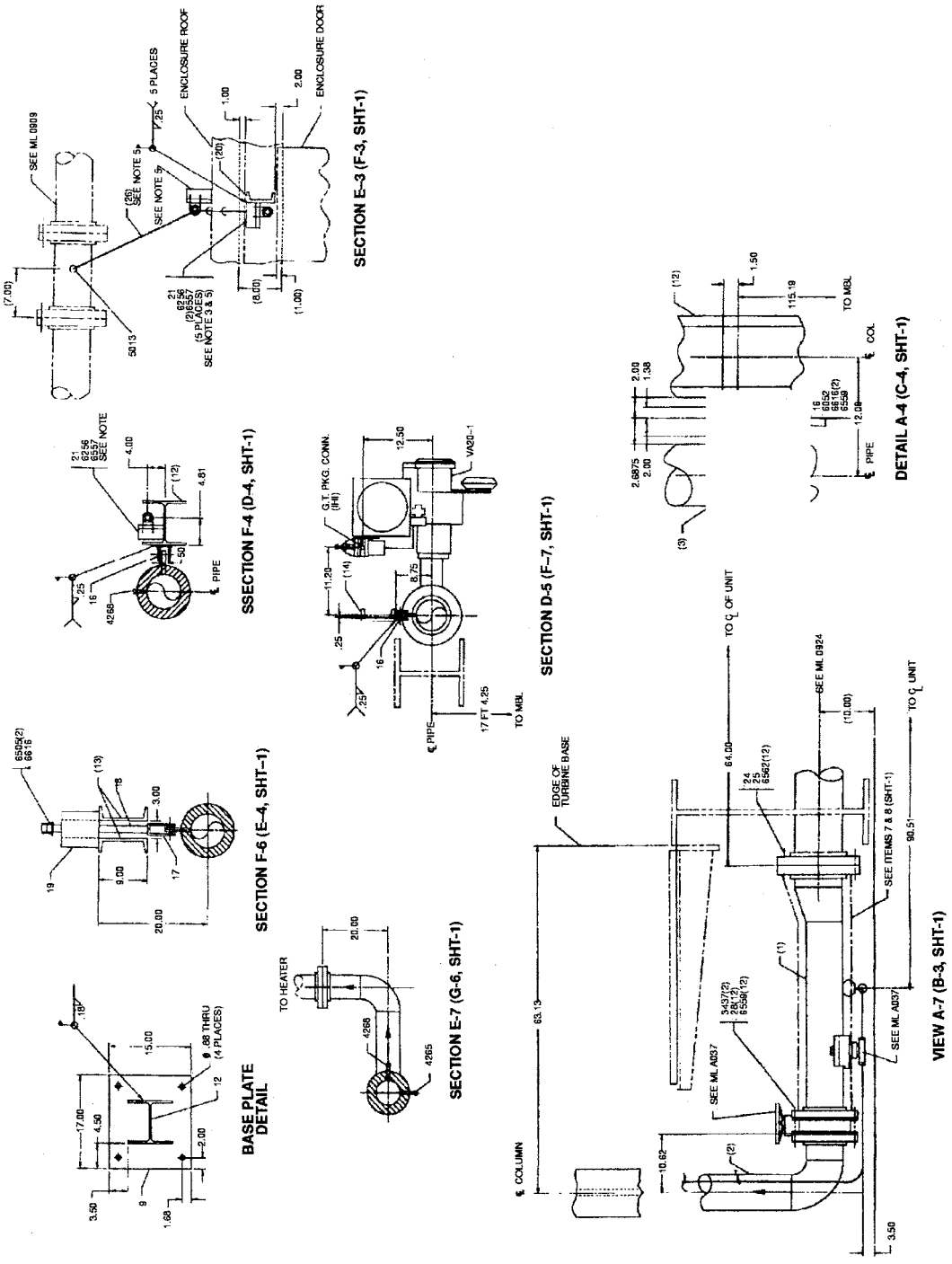


Figure 18. Inlet Heating Piping Arrangement (MLI A053) - Typical Layout